

Astronomy for Middle-School Teachers

Two days for teachers of classes 6–8. Confront the misconceptions you and your students share, then rebuild the Earth–Sun–Moon system from things you can do in a school ground with a stick.

6	29	11 h 30 min	1 h 30 min	School: Middle (6–8)
SESSIONS	CONCEPTS TAUGHT	TOTAL	OUTDOORS	AUDIENCE

Moon: start when the moon is **2.5–6.5 days old** (🌑 waxing crescent → 🌒 first quarter)

ASSUMED ALREADY KNOWN

Angles & the Protractor

Gravity: Things Fall

Closed under prerequisites: 2 concepts in total are taken as prior knowledge.

1 Day 1 — What we all get wrong

CLASSROOM · 2 H · DAY 1

KIT Projector · Paper, pencils, protractor

L1 Spin & Axis

A spinning top turns around an invisible line — its axis — and the axis itself stays steady while everything else whirls. Earth is a giant top; its axis points at the pole star.

L1 Cardinal Directions

Finding north, south, east and west from the Sun and shadows. Sunrise and sunset directions drift through the year — due east/west only near the equinoxes.

L1 The Sky Dome

The sky looks like a dome overhead: the horizon is its rim, the zenith its top. Everything we observe is a direction on this dome, not a distance.

L1 Day & Night

Earth spins once in about 24 hours, so the Sun appears to rise in the east and set in the west. Day and night are a matter of which side of Earth faces the Sun.

L1 Sun's Daily Arc

The Sun climbs from the eastern horizon, peaks toward the south (from India), and sinks west. Shadow lengths and directions trace this arc.

TRAINER NOTE Open by asking the room to explain the seasons on paper, before any teaching. Collect the answers — most will say 'the Earth is closer in summer'. Keep the papers and hand them back in Day 2's seasons session. This is the most effective eight minutes in the course, and it only works if you do it before you teach anything.

KIT Gnomon / shadow stick · Measuring tape · Paper, pencils, protractor**L1 Light & Shadow**

Light travels in straight lines, so an object blocks it and casts a shadow shaped like itself. Shadow direction and length depend on where the light is — the key to sundials and eclipses.

L1 Reading the Globe

The globe is Earth's honest model: a tilted ball with poles, continents and oceans. Finding India, the poles and the tilt on a globe comes before any talk of rotation or seasons.

L2 Latitude & Longitude

An address grid for the whole planet: latitude counts degrees from the equator, longitude from Greenwich. Bhopal sits near 23°N, 77°E — and your latitude is written in your sky.

L2 Angles on the Sky

Sky positions and sizes are measured in angles: your fist at arm's length spans about 10°, the Moon half a degree. Degrees, arcminutes and arcseconds are the astronomer's ruler.

L2 Earth is a Sphere

Ships sinking hull-first, new stars rising as you travel south, and Earth's always-round shadow on the eclipsed Moon — classical proofs, no spacecraft needed.

L2 Gnomon & Local Noon

A vertical stick tells time: its shadow is shortest at local noon, pointing exactly north–south. The gnomon (shanku) is astronomy's oldest instrument.

L3 Measuring the Earth

Eratosthenes sized the Earth around 240 BCE with two sticks and a noon shadow angle. Two schools at different latitudes can repeat it today over WhatsApp.

TRAINER NOTE A stick in the ground gives local noon, the true cardinal line, the latitude, and — with a partner school a few hundred km north or south — the circumference of the Earth. Teachers who do this once run it every year: no budget, no breakage, and a result the class computes themselves. If you cut one thing from this course, do not cut this.

3 Day 1 — Evening sky: what a beginner actually sees

OBSERVING NIGHT · 1 H 30 MIN · DAY 1

SKY needs **Thin crescent** · Constellations, Bright stars, Horizon / cardinal points, The Moon

KIT Star chart / planisphere · Red torch

L1 Stars Differ

Stars are not identical dots: they differ in brightness and in colour — steely blue Vega, golden Arcturus, red Betelgeuse. These differences are real physics, visible to the naked eye.

L1 Constellations

Star patterns used as sky landmarks — 88 official constellations today, with older patterns in every culture. They are direction labels, not physical groups.

L1 Asterisms

Popular star patterns that aren't official constellations — the Big Dipper is part of Ursa Major, the Summer Triangle spans three constellations. The sky's informal landmarks.

L2 Diurnal Motion

The whole starry sky wheels westward through the night, circling a fixed point — the celestial pole. This is Earth's rotation seen from the inside.

L2 The Pole Star

Polaris (Dhruva) sits nearly at the north celestial pole, so it barely moves and always marks north. Its altitude above the horizon equals your latitude.

TRAINER NOTE A crescent gives earthshine and an easy first target without washing out the constellations. Teach the fist-at-arm's-length = 10° trick here; the pole star lands best right after they have watched everything else visibly wheel around it for twenty minutes. Expect at least one teacher to be surprised it is not the brightest star in the sky.

4 Day 2 — Seasons and the tilt

INDOOR LAB · 2 H · DAY 2

KIT Projector · Paper, pencils, protractor

L2 Seasons

Earth's axis is tilted 23.4° , so through the year each hemisphere leans toward or away from the Sun — changing day length and the Sun's height. Distance from the Sun is not the cause.

L2 Tropics & Polar Circles

The Tropic of Cancer — which crosses India through MP and Gujarat — marks where the noon Sun can stand exactly overhead. These lines are the axial tilt drawn onto the map.

L2 Equinoxes & Solstices

Four turning points of the year: two equinoxes when day equals night, two solstices when the Sun's arc is highest or lowest. They anchor seasons and calendars worldwide.

L2 Climate Zones & Sun Angle

Torrid, temperate and frigid zones follow from one fact: how steeply sunlight lands. Geography's climate map is astronomy's Sun-angle diagram wearing different clothes.

TRAINER NOTE Hand back Day 1's papers first. A globe and one lamp settles it faster than any slide: keep the tilt fixed and walk the globe around the lamp. The distance explanation dies the moment someone notices it would give both hemispheres the same season at once — let a teacher say that out loud rather than saying it yourself.

5 Day 2 — The Moon, slowly

INDOOR LAB · 2 H · DAY 2

KIT Projector · Paper, pencils, protractor

L1 Observing the Moon

The Moon is visible in the day as often as at night, rises about 50 minutes later each day, and shows changing shapes. Just watching it for a month is a complete experiment.

L2 Phases of the Moon

The Moon is always half-lit by the Sun; the phase is how much of that lit half faces us, cycling in about 29.5 days. Phase, rise time and Sun–Moon angle all lock together.

L2 Moon Orbits Earth

The Moon circles Earth roughly monthly at about 384,000 km — around 30 Earth-diameters away. Its orbital motion is what drives the phase cycle and its nightly 13° eastward drift.

L2 Lunar Eclipses

At full moon, the Moon can pass through Earth's shadow and turn copper-red — sunset light bent through our atmosphere. Safe to watch, visible from the whole night side.

L2 Solar Eclipses

At new moon, the Moon's small shadow can sweep across Earth, briefly hiding the Sun. Totality happens because the Moon and Sun span the same half-degree — a cosmic coincidence.

TRAINER NOTE Ball-and-lamp, one ball per teacher, everyone holding their own moon at arm's length and turning on the spot. Phases are a viewing-angle effect and the room has to see it from inside the orbit, not from a diagram drawn above it. Only then contrast with an eclipse — the shadow explanation of phases is the single misconception this session exists to kill, and a diagram will not kill it.

6 Day 2 — Building your own lesson

HANDS-ON / BUILD · 1 H 30 MIN · DAY 2

KIT Paper, pencils, protractor

L1 Maps & Scale Models

A map shrinks the real world by a fixed scale — 1 cm standing for 1 km — keeping shapes and directions honest. Star charts, globes and solar-system walks are all the same idea.

L2 Models in Science

A globe, a map, a diagram of the Sun — none are the real thing; all are models that answer some questions well and others not at all. Science improves by testing models against observation.

L2 Observation vs Inference

'The pavement is wet' is an observation; 'it rained last night' is an inference — a story you added. Science advances by keeping the two on separate lines of the notebook, because inferences can be wrong while the observation stays true, and a lab report that blurs them cannot be checked by anyone.

TRAINER NOTE Each teacher leaves with one lesson plan, for one concept from this course, using only what their school already owns. Peer-review in pairs against the misconception list — the test of a plan here is whether it would have caught the teacher's own Day 1 answer.

FULL KIT LIST

Gnomon / shadow stick <i>(consumable)</i>	sessions 2
Measuring tape	sessions 2
Projector	sessions 1, 4, 5
Red torch	sessions 3
Star chart / planisphere	sessions 3
Paper, pencils, protractor <i>(consumable)</i>	sessions 1, 2, 4, 5, 6